14. Extending Zoho Creator with Pro-Code Features Using Zoho Catalyst and Custom Functions

Aim

To enhance a Zoho Creator application by integrating custom Deluge functions and Zoho Catalyst serverless services to perform advanced calculations, connect external APIs, and handle complex data-processing tasks.

Theory

Zoho Creator is a low-code platform that enables rapid application development using forms, reports, workflows, and Deluge scripting. For advanced requirements, developers can extend its capabilities using custom functions and external serverless services.

1. Custom Deluge Functions

Custom Deluge functions allow developers to create reusable scripts for specific business operations within Zoho Creator.

Advantages:

- Automates repetitive business tasks.
- Executes calculations and validations.
- Connects with external services through APIs.
- Reduces duplicate code by reusing functions across workflows.

2. Zoho Catalyst Integration

Zoho Catalyst is a serverless development platform that supports pro-code application development. It can be used to create backend services that handle specialized processing outside the main Zoho Creator application.

Advantages:

- Supports backend development using languages such as Java, Node.js, and Python.
- Helps separate complex processing from low-code workflows.
- Enables integration with external APIs and services.

- Provides scalable serverless components, subject to platform limits and configuration.

Procedure

Step 1: Create a Custom Function in Zoho Creator

1. Log in to Zoho Creator and open the required application.
2. Navigate to the application's workflow or logic settings.
3. Open the Functions section and create a new function.
4. Specify the function name, input parameters, and return type.
5. Write the Deluge script to perform the required operation, such as calculating discounts or validating order details.
6. Save the function and associate it with the appropriate form workflow.

Step 2: Set Up a Zoho Catalyst Project

1. Sign in to the Zoho Catalyst Console.
2. Create a new project for the backend service.
3. Choose a supported programming language, such as Node.js or Python.
4. Develop a serverless function to perform a task such as generating reports, processing data, or executing complex calculations.
5. Test the function to ensure that it produces the expected output.

Step 3: Connect Zoho Creator with Zoho Catalyst

1. Deploy the Catalyst function using the appropriate deployment method.
2. Configure the required authentication and API access permissions.
3. Create a secure connection in Zoho Creator for communicating with the external service.
4. Use Deluge's `invokeurl` task to send data from the Creator application to the Catalyst endpoint.
5. Receive the response and process the returned information within the application.

Example Deluge code:

```
response = invokeurl
[
   url: "YOUR_CATALYST_API_ENDPOINT"
   type: POST
   parameters: {"recordId": input.ID}
   connection: "catalyst_connection"
];
info response;
```

Note: Replace the endpoint and connection name with the actual configured values. The request format must match the deployed API's requirements.

Step 4: Test and Verify the Integration

1. Open the relevant form in Zoho Creator.
2. Enter sample data and submit the form.
3. Verify that the custom Deluge function executes correctly.
4. Check whether the data reaches the Catalyst backend service.
5. Confirm that the backend processes the request and returns the expected response.
6. Verify the final output in the Creator application and review any errors.

Result

The Zoho Creator application was enhanced using custom Deluge functions and Zoho Catalyst integration. The custom functions automated business operations, while the Catalyst backend provided additional capabilities for complex processing and external API integration. This approach improved modularity, flexibility, and the potential scalability of the application.